

Total No. of Questions : 8]

SEAT No. :

PC-2847

[Total No. of Pages : 2

[6352]-71

S.E. (Artificial Intelligence & Machine Learning)
OBJECT ORIENTED PROGRAMMING
(2019 Pattern) (Semester-III) (218544)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data, if necessary.*

- Q1)** a) What is Destructors? Explain finalize function. [6]
b) Write a program to calculate an area of any polygon using parameterized constructor. [6]
c) Explain different types of Constructors in OOP. [6]

OR

- Q2)** a) Explain concept of symbolic constant with example. [6]
b) Differentiate between Constructor and Destructor. [6]
c) Write Pseudo code for Constructor Overloading. [6]

- Q3)** a) Write program to implement hierarchical inheritance. [5]
b) Explain Method overriding with example. [6]
c) Define Inheritance. What are the types of Inheritance? [6]

OR

- Q4)** a) Write a program for addition of two numbers using Method overloading. [5]
b) What is abstract class? Explain with example. [6]
c) What is interface? How to declare an interface, write syntax? Can we achieve multiple inheritance by using interface? [6]

P.T.O.

- Q5)** a) What is Run Time Exception? Demonstrate with the example, how to handle the Number Format Exception? [6]
b) Explain try, catch and finally with suitable example. [6]
c) What is Collection? Explain set interface? [6]

OR

- Q6)** a) What is checked exception? Demonstrate Arithmetic Exception using throw. [6]
b) What is Generics? Write a code of Array List using generics. [6]
c) Differentiate between Array List and Linked List collection classes. [6]

- Q7)** a) Explain Concept of Stream with its types. [6]
b) Explain singleton pattern with example. [6]
c) Write the program to read a file using FileInputStream class. [5]

OR

- Q8)** a) Explain Iterator Pattern with its merits. [6]
b) Write a program to write a file using File Output Stream class. [6]
c) What are design patterns? Explain 'Adapter' design pattern in detail. [5]

